CareBridge Health

Digital Referral Management Platform

SOFTWARE REQUIREMENTS SPECIFICATION

SRS Document — Version 1.0

Document Type	Software Requirements Specification
Project	CareBridge Health Platform
Technology Stack	MERN Stack (MongoDB, Express.js, React.js, Node.js)
Target Market	Pakistan Healthcare Market (PKR-based)
Version	1.0 — Initial Release

CONFIDENTIAL — CareBridge Health Pakistan

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1. Introduction & Project Overview

1.1 Purpose

This Software Requirements Specification (SRS) document defines the complete functional and non-functional requirements for CareBridge Health — a three-tier digital referral management platform built on the MERN stack. It serves as the authoritative reference for development, QA, and stakeholder review.

1.2 Project Background

Pakistan's healthcare ecosystem currently relies on informal referral channels — primarily WhatsApp messages and personal contacts — to route patients from consulting physicians to hospitals. This approach results in:

- Untracked patient flow and revenue leakage for hospitals
- No structured follow-up or accountability mechanism
- Consultants unable to monitor referral outcomes or earnings
- 60–70% of hospital patient flow arriving via referrals with zero centralized tracking

CareBridge Health replaces this informal network with a structured, trackable, and revenue-generating digital platform.

1.3 Product Vision

CareBridge Health connects three core stakeholder groups on a single platform: Consultants (referring doctors), Hospitals (receiving facilities), and Admins (platform managers). The platform automates referral routing using an AI-assisted scoring engine, tracks patient journeys end-to-end, and provides full PKR-based financial visibility to all parties.

1.4 Scope

Module	Scope Description
Consultant Module	Referral initiation, patient intake, smart hospital suggestions, earnings, referral tracking
Hospital Module	Referral inbox, bed management, department routing, analytics, PKR billing
Admin Module	User management, platform analytics, scoring configuration, reports
Payment Module	PKR payment tracking, JazzCash/EasyPaisa integration, payout processing

1.5 Definitions & Acronyms

Term	Definition
MERN	MongoDB + Express.js + React.js + Node.js technology stack
PKR	Pakistani Rupee — all financial values are denominated in PKR

SLA	Service Level Agreement — response time thresholds per urgency level
PMDC	Pakistan Medical & Dental Council — physician registration number
ICU/NICU/PICU	Intensive Care / Neonatal ICU / Pediatric ICU bed categories
JWT	JSON Web Token — authentication mechanism
API	Application Programming Interface — backend service endpoints

2. System Architecture & Technology Stack

2.1 High-Level Architecture Diagram

— PRESENTATION LAYER (React.js) —
Consultant Portal | Hospital Admin Panel | Super Admin Dashboard
— API / BUSINESS LOGIC LAYER (Node.js + Express.js) —
Auth Service | Referral Engine | Scoring Algorithm | Notification Service | Payment Service
— DATA LAYER (MongoDB + Redis) —
MongoDB Atlas (Primary DB) | Redis (Caching/Real-time) | Cloudinary (File Storage)
— EXTERNAL INTEGRATIONS —
JazzCash / EasyPaisa | Firebase (Push Notifications) | Google Maps API | Twilio (SMS)

2.2 Three-Tier User Roles

Role	Portal	Primary Responsibility
Consultant	Consultant Web & Mobile App	Initiate referrals, track outcomes, receive PKR payouts
Hospital Admin	Hospital Admin Dashboard	Manage referral inbox, bed inventory, billing, analytics
Super Admin	Admin Control Panel	Platform management, user approval, scoring config, reports

2.3 Communication & Real-Time

- REST APIs over HTTPS for all CRUD operations
- WebSocket (Socket.io) for real-time referral status updates and bed availability changes
- Firebase Cloud Messaging (FCM) for push notifications to mobile clients
- Redis Pub/Sub for broadcasting live bed status changes across hospital dashboard

3. Functional Requirements

3.1 Authentication & User Management

ID	Requirement	Description
FR-01	Role-based registration	Platform supports 3 roles: Consultant, Hospital Admin, Super Admin. Each role has a distinct registration flow.
FR-02	PMDC Verification	Consultant registration requires PMDC number. Admin manually verifies before account activation.
FR-03	JWT Authentication	All API endpoints protected by JWT tokens. Access tokens expire in 15 minutes; refresh tokens in 7 days.
FR-04	Password Reset	Email-based OTP for password reset. SMS fallback via Twilio.
FR-05	Profile Management	All users can update their profile, contact details, and preferences.

3.2 Consultant Module Requirements

ID	Requirement	Description
FR-10	Smart Intake Form	Search by Symptoms, Department, Doctor, Procedure, Budget. Auto-detect department from symptoms using keyword mapping.
FR-11	Patient Details Capture	Capture Name, Age, Gender, Phone, Area/Locality, CNIC (optional).
FR-12	Clinical Details Entry	Searchable symptom tags, provisional diagnosis, urgency (Emergency/Urgent/Routine), clinical notes, file upload (PDF/image).
FR-13	Smart Hospital Suggestions	Display top 5 hospital cards ranked by 6-factor scoring. Show name, distance, bed availability, ICU status, PKR cost range, rating.
FR-14	Referral Submission	Review summary, obtain patient consent checkbox, submit with generated Referral ID and Promo Code.
FR-15	My Referrals Dashboard	Filter by status (Pending/Accepted/Rejected/Admitted/Closed), date range, specialty, hospital.
FR-16	Earnings & Statements	Case-wise PKR payouts, monthly totals, downloadable PDF statements, promo code use notifications.

3.3 Hospital Admin Module Requirements

ID	Requirement	Description
FR-20	Live Dashboard	Real-time widgets: Total Referrals Today, Pending Referrals, Admissions, Revenue, Bed Status Snapshot.

FR-21	Live Bed Management	Per-ward bed count management with +/- controls. Wards: General, Private, ICU, NICU, PICU. Real-time sync.
FR-22	Referral Inbox	Centralized inbox with Accept/Reject workflow. SLA countdown displayed. Rejection requires documented reason.
FR-23	Department Routing	Incoming referrals auto-routed to relevant department head based on specialty. Manual override available.
FR-24	Rate Package Manager	Define PKR Price-Min/Max per service per department. Effective date tracking for historical billing.
FR-25	Reports & Analytics	Referral conversion rate, avg response time, dept-wise admissions, monthly PKR billing totals, KPI metrics.

3.4 Admin Module Requirements

ID	Requirement	Description
FR-30	User Approval Workflow	Review and approve/reject Consultant and Hospital registrations. PMDC verification, document review.
FR-31	Platform Analytics	Cross-platform KPIs: total referrals, acceptance rates, revenue by hospital, consultant performance.
FR-32	Scoring Configuration	Admin can adjust weights of the 6-factor scoring formula without code deployment.
FR-33	Specialty & Tag Master	Manage department list, symptom tag library, procedure catalog used across platform.
FR-34	Payout Management	Trigger and track consultant PKR payouts. Monthly reconciliation with downloadable reports.

4. Application Flow & Navigation

4.1 End-to-End Referral Lifecycle

Step	Actor	Action
1	**Consultant**	Logs into Consultant Portal → Opens New Referral → Fills patient info, symptoms, urgency, uploads reports
2	**System (Auto)**	Scoring Engine activates → Maps symptoms to departments → Fetches active hospitals with matching dept + available beds → Applies 6-factor score → Returns Top 5 hospital cards
3	**Consultant**	Reviews suggested hospitals → Selects target hospital → Reviews referral summary → Patient gives consent → Clicks Submit
4	**System (Auto)**	Generates unique Referral ID + Promo Code → Sends real-time notification to Hospital Admin → Starts SLA countdown timer
5	**Hospital Admin**	Receives referral in inbox → Reviews patient details and clinical info → Accepts or Rejects within SLA window
5a	**System (SLA Breach)**	If no response within SLA → Auto-escalates to next-ranked hospital in the Top 5 queue → Repeat until accepted
6	**Hospital Admin**	Schedules appointment → Assigns bed → Notifies consultant and patient of acceptance
7	**Patient (Physical)**	Arrives at hospital → Front desk verifies Referral ID → Tags as 'CareBridge Referral' in hospital system
8	**Hospital Admin**	Records all services under Referral ID → Generates final PKR bill → Records payment (JazzCash/EasyPaisa/Cash) → Confirms on platform
9	**System (Auto)**	Case marked Completed → Triggers consultant payout calculation → Updates monthly PKR reconciliation → Refreshes all dashboards

4.2 SLA Response Time Rules

Urgency Level	SLA Window	Auto-Escalation Trigger
EMERGENCY	Immediate (< 15 min)	After 15 minutes of no response, escalate to next hospital
URGENT	≤ 2 hours	After 2 hours of no response, escalate to next hospital
ROUTINE	≤ 24 hours	After 24 hours of no response, escalate to next hospital

4.3 Navigation Flow Map

AUTHENTICATION ENTRY POINT

Consultant Login	Hospital Admin Login	Super Admin Login
→ Consultant Dashboard	→ Hospital Dashboard	→ Admin Control Panel

5. Screen Inventory & Screen-by-Screen Specification

5.1 Complete Screen List

SCR#	Screen Name	Module	Primary Purpose
01	**Splash / Landing Screen**	Auth	App entry with login options for each role
02	**Consultant Registration**	Auth	New consultant account with PMDC verification
03	**Hospital Registration**	Auth	New hospital onboarding with document upload
04	**Login Screen**	Auth	Role-based login with JWT authentication
05	**Forgot / Reset Password**	Auth	Email OTP-based password recovery
06	**Consultant Dashboard (Home)**	Consultant	Overview: recent referrals, earnings, quick actions
07	**New Referral — Smart Intake Form**	Consultant	Multi-step intake: search, patient info, clinical details
08	**Hospital Suggestion Cards**	Consultant	Top 5 scored hospital display with send referral button
09	**Referral Confirmation Screen**	Consultant	Summary review + patient consent + submit
10	**My Referrals (Tracking List)**	Consultant	Filtered list of all referrals with real-time status
11	**Referral Detail View**	Consultant	Full timeline of a single referral case
12	**Earnings & Statements**	Consultant	PKR payouts, monthly totals, downloadable statements
13	**Consultant Profile / Settings**	Consultant	Profile update, specialty, clinic info, promo code
14	**Hospital Dashboard (Home)**	Hospital	Live widgets: referrals today, pending, admissions, revenue
15	**Live Bed Management Panel**	Hospital	Ward-wise bed count with +/- live update controls
16	**Referral Inbox**	Hospital	All incoming referrals with Accept/Reject + SLA timer
17	**Referral Detail (Hospital View)**	Hospital	Patient info, clinical details, history for a referral
18	**Admission Management**	Hospital	Record admit date, services, discharge, final PKR bill

19	**Rate Package Manager**	Hospital	Define PKR min-max for each service/department
20	**Hospital Analytics & Reports**	Hospital	Conversion rate, avg response time, dept admissions, KPIs
21	**Hospital Profile / Settings**	Hospital	Hospital info, department config, emergency toggle
22	**Admin Control Panel (Home)**	Admin	Platform-wide stats: users, referrals, revenue overview
23	**User Approval Queue**	Admin	Review & approve/reject consultant and hospital registrations
24	**Consultant Management**	Admin	List, activate, deactivate, view all consultants
25	**Hospital Management**	Admin	List, activate, deactivate, view all hospitals
26	**Platform Analytics Dashboard**	Admin	Cross-platform KPIs, charts, trend analysis
27	**Scoring Formula Config**	Admin	Adjust 6-factor weights for hospital suggestion engine
28	**Specialty & Tag Master**	Admin	Manage departments, symptom tags, procedure catalog
29	**Payout Management**	Admin	Trigger payouts, reconcile PKR, download reports
30	**Notifications Center**	All	Centralized notification feed for all role-based alerts

5.2 Detailed Screen Specifications

SCR-01: Splash / Landing Screen *[Auth Module]*

Purpose	Application entry point. Displays CareBridge branding. User selects their role (Consultant, Hospital Admin) to proceed to the correct registration or login flow.
Key Inputs	Role selection button tap (Consultant / Hospital Admin / Admin)
Key Outputs	Routes user to role-specific login or registration screen
User Role	All users (unauthenticated)

SCR-07: New Referral — Smart Intake Form *[Consultant Module]*

Purpose	Core screen for initiating a patient referral. Multi-step form with smart search. Step 1: Search parameters. Step 2: Patient demographics. Step 3: Clinical details and file uploads.

Key Inputs	Search query (symptoms/department/doctor/procedure/budget), Patient Name, Age, Gender, Phone, Area, CNIC (optional), Symptom tags (multi-select), Provisional diagnosis, Urgency level dropdown, Clinical notes text area, File upload (PDF/JPG)
Key Outputs	Populated intake form submitted to Scoring Engine, displays loading state while engine computes results
User Role	Consultant

SCR-08: Hospital Suggestion Cards *[Consultant Module]*

Purpose	Displays top 5 hospitals ranked by the scoring algorithm. Each card shows all key decision-making data. Consultant reviews and selects the best-fit hospital.
Key Inputs	Tap 'Send Referral' on a hospital card, optional filter/sort controls
Key Outputs	5 hospital cards showing: Hospital name, Distance (km from patient area), Bed availability count, ICU/NICU/PICU status badges, PKR cost range (min–max), Available consultants, Star rating, 'Send Referral' CTA button
User Role	Consultant

SCR-14: Hospital Dashboard (Home) *[Hospital Module]*

Purpose	Live overview for hospital admin. Real-time KPI widgets auto-refresh via WebSocket. Central navigation hub for all hospital operations.
Key Inputs	No primary user input — reads from live data stream
Key Outputs	Widgets: Total Referrals Today (count), Pending Referrals (count with alert if SLA approaching), Admissions from Platform (today), Revenue from Platform Cases (PKR), Bed Status Snapshot (% occupancy by ward), Quick navigation to Inbox, Beds, Analytics
User Role	Hospital Admin

SCR-15: Live Bed Management Panel *[Hospital Module]*

Purpose	Allows hospital admin to update bed availability in real time. Changes immediately propagate to the Scoring Engine so that referral suggestions reflect accurate data.
Key Inputs	+ / − increment buttons per ward row, Emergency enabled toggle
Key Outputs	Table rows for: General Ward, Private Ward, ICU, NICU, PICU. Columns: Ward Name, Total Beds, Occupied, Available (highlighted in green/red based on availability). Changes broadcast via WebSocket.
User Role	Hospital Admin

SCR-16: Referral Inbox *[Hospital Module]*

Purpose	Centralized inbox for all incoming referral requests. Displays SLA countdown timers. Each referral card shows urgency level with color coding. Auto-routes to department head.
Key Inputs	Accept button, Reject button (requires reason text input), Department assignment dropdown
Key Outputs	List of referral cards sorted by urgency. Each card shows: Referral ID, Patient initials, Age, Gender, Urgency badge (color-coded), Specialty, Referring consultant name, SLA countdown timer, Accept / Reject action buttons
User Role	Hospital Admin / Department Head

SCR-22: Admin Control Panel (Home) *[Admin Module]*

Purpose	Super admin overview of the entire platform. Aggregated metrics across all hospitals and consultants. Entry point to all admin sub-modules.
Key Inputs	Navigation clicks to sub-modules
Key Outputs	Widgets: Total Registered Users, Pending Approvals (count), Total Referrals Processed (all time), Active Hospitals, Active Consultants, Platform Revenue (PKR), Recent activity log, Quick action buttons
User Role	Super Admin

6. MERN Stack Technology Distribution

6.1 Technology Map by Layer

M — MongoDB (Database Layer)

Component	Usage in CareBridge
MongoDB Atlas	Cloud-hosted MongoDB cluster. All primary application data: users, referrals, hospitals, beds, admissions, payments.
Collections	users, consultants, hospitals, departments, beds_inventory, rate_packages, referrals, admissions, payouts, notifications, symptom_tags
Mongoose ODM	Schema validation, relationships via population (virtuals + refs), middleware hooks for pre-save operations (e.g., hashing passwords).
Geospatial Indexing	2dsphere index on hospitals.location for distance-based queries (used in scoring algorithm's distance factor).
Text Indexing	Full-text search indexes on symptom_tags, departments, and doctor names to power the smart intake form search.
Aggregation Pipeline	Complex analytics queries: monthly revenue totals, referral conversion rates, consultant performance ranking.
Redis (Cache)	Caches top-5 hospital suggestions per intake combination (TTL: 5 min). Pub/Sub for live bed update broadcasts.

E — Express.js (API / Routing Layer)

Component	Usage in CareBridge
RESTful Router	Organized route modules: /auth, /consultants, /hospitals, /referrals, /beds, /payments, /admin, /notifications.
Middleware Stack	JWT authentication middleware, role-guard middleware (consultant/hospital/admin), request validation (Joi/express-validator), error handler.
Multer	Handles multipart file uploads from the intake form (lab PDFs, images). Files streamed to Cloudinary.
Scoring Engine Service	Dedicated Express service module: fetches hospitals, runs 6-factor score calculation, returns ranked results.
Socket.io (Server)	WebSocket server integrated into Express HTTP server. Handles real-time channels: referral-status, bed-updates, notifications.
Cron Jobs (node-cron)	Scheduled jobs for SLA monitoring (checks every minute for timeout referrals), payout calculations (monthly), report generation.

Helmet / CORS	Security middleware: HTTP headers hardening (Helmet), Cross-Origin policy for frontend domain.

R — React.js (Frontend Layer)

Component	Usage in CareBridge
Create React App / Vite	Project scaffolding. Vite recommended for faster build/HMR. Separate apps or lazy-loaded routes per role.
React Router v6	Nested routing per role. Protected routes using JWT context. Role-specific layout wrappers.
Redux Toolkit	Global state: auth (user/token), referral list, notifications counter, bed status, active filters.
React Query (TanStack)	Server state management for all API calls: caching, background refetch, optimistic updates on Accept/Reject actions.
Ant Design / MUI	Component library for dashboards, tables, forms, modals, cards. Customized to CareBridge branding.
Socket.io (Client)	Subscribes to referral-status and bed-update channels. Dispatches to Redux on event receipt.
Recharts / Chart.js	Charts for analytics screens: line (monthly revenue), bar (referral volume), donut (bed occupancy).
React Hook Form	Multi-step intake form management with Yup schema validation. File input integration with progress indicator.
Google Maps React	Hospital suggestion cards show distance. Map view available for hospital location visualization.

N — Node.js (Runtime Layer)

Component	Usage in CareBridge
Node.js v18+ LTS	Application runtime. Async I/O handles high-concurrency referral submissions and real-time WebSocket connections.
bcryptjs	Password hashing for user accounts. Salt rounds = 12.
jsonwebtoken	JWT generation and verification. Separate secret keys for access tokens and refresh tokens.
Nodemailer	Transactional emails: registration OTP, referral status updates, payout confirmation.
Twilio SDK	SMS notifications to consultants and patients for referral acceptance and appointment confirmation.
Firebase Admin SDK	Server-side push notification dispatch to mobile clients (consultants using the mobile app).
Cloudinary SDK	Uploads and retrieves patient reports, lab PDFs, and hospital documents.

PM2	Production process manager. Cluster mode for multi-core utilization. Zero-downtime restarts.

7. Database Design

7.1 Entity Relationship Overview

The database is MongoDB document-based. Key relationships are maintained via ObjectId references (similar to foreign keys) with Mongoose populate for joined queries.

Core Collection Relationships

Collection A	Collection B	Relationship
users	consultants	One-to-One: Each consultant record links to one user account (userId FK)
users	hospitals	One-to-One: Each hospital record links to one user account (userId FK)
hospitals	beds_inventory	One-to-Many: A hospital has multiple bed entries per ward/department
hospitals	rate_packages	One-to-Many: A hospital defines multiple rate packages per service
consultants	referrals	One-to-Many: A consultant submits multiple referrals
referrals	admissions	One-to-One: An accepted referral results in at most one admission record
admissions	payouts	One-to-One: A completed admission triggers one consultant payout calculation

7.2 Collection Schema Definitions

Collection: users

Field	Type	Notes
_id	ObjectId	Primary Key (auto-generated)
role	String (enum)	consultant | hospital | admin
name	String	Full display name
email	String	Unique, indexed, lowercase
phone	String	Pakistani format (+92...)
passwordHash	String	bcrypt hash, not returned in queries
status	String (enum)	pending | active | suspended
createdAt	Date	Mongoose timestamp
updatedAt	Date	Mongoose timestamp

Collection: referrals

Field	Type	Notes
_id	ObjectId	Primary Key
referralCode	String	Auto-generated unique code (e.g., CB-2024-0001)
consultantId	ObjectId (ref)	→ consultants collection
patientName	String	Patient full name
age	Number	Patient age
gender	String (enum)	male | female | other
phone	String	Patient contact
area	String	Karachi locality / town
cnic	String	Optional, encrypted at rest
urgency	String (enum)	emergency | urgent | routine
symptomsText	String	Free text + stored tags
symptomTags	[ObjectId]	Refs to symptom_tags collection
diagnosisText	String	Provisional diagnosis
notes	String	Additional clinical notes
attachments	[String]	Cloudinary URLs for uploaded files
targetHospitalId	ObjectId (ref)	→ hospitals collection
status	String (enum)	pending | accepted | rejected | admitted | closed
promoCode	String	Consultant's promo code assigned at submission
scoringData	Object	Snapshot of scoring factors at time of referral
createdAt	Date	Submission timestamp

8. Smart Hospital Scoring Engine

8.1 Algorithm Overview

The Scoring Engine is a server-side module that executes on every referral intake submission. It evaluates all active hospitals against 6 weighted factors and returns the top 5 ranked results in under 2 seconds.

8.2 Scoring Formula

#	Factor	Weight	Calculation Logic
1	**Specialty Match**	**30%**	Binary or partial match: does the hospital's active departments include the required specialty? Full match = 30, partial = 15.
2	**Bed Availability**	**25%**	Score = (available beds / total beds) × 25. Zero available beds = 0 score → hospital excluded from suggestions.
3	**Distance from Patient Area**	**15%**	MongoDB $geoNear query. Score = 15 × (1 - distance/max_radius). Closer hospitals score higher.
4	**Cost Fit (Budget Range)**	**10%**	Hospital's PKR rate package range vs consultant's specified budget. Overlap score: full = 10, partial = 5, none = 0.
5	**Acceptance Speed / SLA History**	**10%**	Historical average response time for this hospital vs SLA. Faster historical acceptance = higher score (0–10).
6	**Preferred Hospital / Doctor**	**10%**	Consultant's preference list from past referrals or manually saved preferences. Preferred = +10 bonus.
∑	**TOTAL SCORE**	**100%**	Max score = 100. Top 5 hospitals by score are returned. Hospitals with 0 beds are excluded before scoring.

8.3 Engine Execution Pipeline

1. Receive intake payload (symptoms, area coordinates, urgency, budget, specialty)
2. If department not specified: map symptom tags → department via keyword-to-dept lookup table
3. Query MongoDB for hospitals with: status=active AND matching department AND beds_available > 0
4. For each candidate hospital: compute 6-factor scores and sum to total_score
5. Sort candidates descending by total_score
6. Return top 5 results with score breakdown (for transparency display on suggestion cards)
7. Cache result in Redis with key = hash(intake_params), TTL = 5 minutes

9. Non-Functional Requirements

ID	Category	Requirement
NFR-01	**Performance**	Scoring Engine must return top 5 results within 2 seconds. API response time P95 < 500ms under 500 concurrent users.
NFR-02	**Real-Time Latency**	Bed status updates must reflect on all connected dashboards within 1 second of change.
NFR-03	**Availability**	Platform uptime SLA: 99.5% monthly. Scheduled maintenance windows communicated 48 hours in advance.
NFR-04	**Scalability**	Architecture must support horizontal scaling. MongoDB Atlas auto-scaling enabled. PM2 cluster mode on Node.js.
NFR-05	**Security**	All data in transit over TLS 1.2+. Patient CNIC encrypted at rest (AES-256). JWT secret rotation every 90 days.
NFR-06	**Data Privacy**	Patient data access restricted to: the referring consultant and the accepted hospital only. Admin access logged.
NFR-07	**Usability**	Intake form completion time target: under 3 minutes for a standard referral. Mobile-responsive design mandatory.
NFR-08	**Localization**	All PKR values formatted with PKR prefix and comma-separated thousands (e.g., PKR 1,50,000). Urdu language support in Phase 2.
NFR-09	**Audit Trail**	All referral status changes, bed updates, and payout events logged with actor ID, timestamp, and IP address.
NFR-10	**Backup & Recovery**	MongoDB Atlas automated daily backups with 7-day retention. Point-in-time recovery within 4-hour RTO.

10. API Design Overview

10.1 API Conventions

- Base URL: https://api.carebridge.health/v1/
- All responses in JSON format with envelope: { success, data, message, pagination? }
- Authentication: Bearer {JWT_ACCESS_TOKEN} in Authorization header
- Pagination: ?page=1&limit=20 for list endpoints
- Date format: ISO 8601 (2024-01-15T10:30:00Z)

10.2 Core API Endpoints

Method	Endpoint	Purpose	Auth Required
POST	/auth/register	Register new user (consultant / hospital)	None
POST	/auth/login	Login, returns access + refresh tokens	None
POST	/referrals	Submit new referral intake form	Consultant JWT
GET	/referrals/suggestions	Get scored top-5 hospital suggestions	Consultant JWT
GET	/referrals/:id	Get referral details & timeline	Consultant / Hospital JWT
PATCH	/referrals/:id/accept	Hospital accepts referral	Hospital JWT
PATCH	/referrals/:id/reject	Hospital rejects with reason	Hospital JWT
GET	/hospitals/:id/beds	Get live bed status for a hospital	Hospital JWT
PATCH	/hospitals/:id/beds/:ward	Update bed count for a ward	Hospital JWT
GET	/consultants/earnings	Get earnings summary and case payouts	Consultant JWT
GET	/admin/analytics	Platform-wide analytics and KPIs	Admin JWT

11. Security & Compliance

11.1 Authentication & Authorization

- JWT-based stateless authentication with short-lived access tokens (15 min) and rotating refresh tokens (7 days)
- Role-based access control (RBAC): consultant, hospital, admin — enforced at middleware level
- All passwords hashed with bcrypt, salt rounds = 12
- Rate limiting on auth endpoints: max 10 requests/minute per IP to prevent brute force

11.2 Data Security

- All API communication over TLS 1.2+
- Patient CNIC field encrypted at rest using AES-256
- Uploaded medical documents stored in Cloudinary with private access URLs (signed, time-limited)
- MongoDB Atlas network access restricted to application server IPs

11.3 Compliance Considerations

- SHCC (Sindh Healthcare Commission) compliance guidelines followed for data handling
- Patient data access limited to the referring consultant and the accepting hospital only
- Full audit log maintained for all data access and mutations on patient records
- Admin access to patient records requires documented justification in the audit log

12. Payment Integration (PKR)

12.1 Supported Payment Channels

Channel	Type	Implementation
JazzCash	Mobile Wallet (PKR)	JazzCash REST API integration for online PKR transactions with auto-reconciliation per referral case.
EasyPaisa	Mobile Wallet (PKR)	EasyPaisa API for digital payment capture. Webhook-based confirmation triggers payout calculation.
Bank Transfer	Direct Bank (PKR)	Manual entry of transaction reference by hospital admin. Requires admin confirmation before payout trigger.
Cash	Cash Receipt (PKR)	Cash receipt logging with manual entry by hospital. Amount confirmed by hospital admin, tracked under Referral ID.

12.2 PKR Payout Logic

- Consultant payout is triggered when: Admission is marked complete AND final PKR bill is confirmed
- Payout amount calculated based on agreed rate per referral type (defined in admin payout config)
- Monthly reconciliation report generated on the 1st of each month for the previous month
- Consultants can download PDF statements from the Earnings screen
- All PKR values stored as integers (in paisa) to avoid floating-point errors

13. Glossary

Term	Definition
CareBridge Health	Three-tier digital referral management platform for the Pakistani healthcare market
Consultant	A licensed medical doctor or physician who refers patients to hospitals via the platform
Hospital Admin	The designated user account for a hospital facility managing its referral inbox and bed data
Super Admin	Platform operator with full access to all modules, configurations, and analytics
Referral	The digital record of a doctor recommending a patient to a specific hospital for treatment
Referral ID	Unique system-generated identifier for each referral (format: CB-YYYY-NNNN)
Promo Code	Unique code assigned to each consultant, attached to their referrals for tracking and payout attribution
Scoring Engine	The algorithmic module that ranks hospitals based on 6 weighted factors for each referral intake
SLA	Service Level Agreement — the maximum allowed response time for a hospital to accept/reject a referral
Auto-Escalation	Automatic forwarding of a referral to the next-ranked hospital when the current hospital's SLA window expires
Urgency Level	Classification of referral priority: Emergency (immediate), Urgent (≤2 hrs), Routine (≤24 hrs)
Bed Inventory	Real-time record of total, occupied, and available beds per ward per hospital
Rate Package	Defined PKR price range (min–max) for a specific medical service at a hospital
Payout	PKR payment disbursed to a consultant upon successful completion of a referred patient case
MERN Stack	MongoDB + Express.js + React.js + Node.js — the full-stack JavaScript technology combination used in this project
WebSocket	Persistent two-way communication protocol used for real-time updates (bed status, referral notifications)
JWT	JSON Web Token — compact, URL-safe means of representing authentication claims between parties
PKR	Pakistani Rupee — the currency used for all financial transactions on the platform
PMDC	Pakistan Medical and Dental Council — the regulatory body issuing license numbers to physicians

Redis	In-memory data store used for caching scoring results and broadcasting real-time bed updates

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END OF SOFTWARE REQUIREMENTS SPECIFICATION

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